Laird TECHNOLOGIES

Innovative **Technology** for a **Connected** World

# Tputty™ 506 Series
## Gap Filler Material




## SOFT, SILICONE PUTTY

Tputty™ 506 is a soft, single part, silicone putty thermal gap filler in which no cure is required. This gap filler is ideal for applications where large gap tolerances are present and in which traditional gap filler pads may apply added pressure on components. This material can be dispensed to fill large and uneven gaps in assemblies.

Tputty™ 506 has a composition which yields superior thermal performance and super compliancy. This material transfers little to no pressure between interfaces. Because Tputty™ 506 is more viscous than grease, bleed and pump-out usually associated with grease is eliminated. Specialized rheology allows for easy flow under pressure. Tputty™ 506 is non-abrasive which leads to less wear on dispensing equipment and therefore reduced equipment maintenance/repair costs.

## FEATURES AND BENEFITS

- Soft and compliant transferring little to no pressure between interfaces
- Non-abrasive
- 3.5 W/mK thermal conductivity
- Available in 75 cc, 180 cc, 360 cc and 600 cc dispensing cartridges
- Available in 20 kg pails
- Easily dispensable from an EFD dispensing system

## APPLICATIONS

- Microprocessors
- Graphic chips
- Automotive
- LED lighting

| | Tputty™ 506 | TEST METHOD |
|---|---|---|
| Construction & Composition | Fully Cured Ceramic-filled dispensable silicone putty | |
| Color | Turquoise | Visual |
| Flow Rate (75 cc taper tip, 0.125" orifice, 90 psi) | 17.2 cc/min | Laird Technologies |
| Temperature Range | -45°C to 200°C | |
| Thermal Conductivity | 3.5 W/mK | Hot Disk |
| Abrasiveness of Predominant Filler | 2 | Mohs hardness scale |
| Density | 1.71 g/cc | He Pycnometer |
| Volume Resistivity | 1.8 x $10^{14}$ Ohm-cm | ASTM D2240 |
| Minimum Bondline thickness mm (in) | 0.1 (0.004) | ASTM D575 |
| Outgassing TML, weight percent | 0.46% | ASTM E595 |
| Outgassing TML, volume percent | 0.79% | ASTM E595 |
| Coefficient of Volumetric Expansion (CVE) | 680 ppm/K | Volume Dilatometry |
| Coefficient of Thermal Expansion (CTE) | 227 ppm/K | Volume Dilatometry |
| Tg | < -90°C | ASTM E1356 |
| Dielectric Constant (1 mHz) | 5.3 | ASTM D150 |
| Specific Heat | 0.85 J/gK | ASTM E1269 |
| Flammability | V0 | UL 94 |

Data for design engineer guidance only. Observed performance varies in application. Engineers are reminded to test the material in application.

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